

QuackTrack

Clean the Mess. Play the Music. Look Ridiculous. Feel Amazing.

Chindogu Group Project

Arduino Powered

Pet-Safe Design

ABOUT THIS PROJECT

What is Chindogu?

The Japanese Art of Useless Inventions

Chindogu is the practice of inventing
gadgets
that solve a real problem in a way so
absurd
that no one would use them in public.

The Group Activity

Students identify a real user
problem, conduct research,
ideate using design thinking,
then prototype a playful
'almost-useless' gadget.

Key Rules of Chindogu

Must solve a REAL problem.
Must be hilariously impractical.
Cannot be sold or patented.
Must cause mild social
embarrassment.

The Problem Statement

Homeowners and pet owners consistently experience gross hair buildup on furniture, clothes and floors during daily cleaning — because standard vacuums can't pick up "stuck" hair, leading to frustrating and unhygienic manual scrubbing.

Who is affected?

Homeowners and pet owners who live with cats, dogs, or long-haired family members

The Pain Point

Standard vacuums fail — manual scrubbing is frustrating and unhygienic

Where it happens

Furniture (sofas), clothes while petting, floors & carpets, washbasins after combing

The Design Brief

The Brief

Design a simple, hands-free tool that quickly removes stuck hair from floors, clothes and furniture — without the user ever having to touch the mess.

Hands-Free

User never needs to touch or handle the hair directly

Quick

Removes hair rapidly without repeated passes or effort

Multi-Surface

Works on floors, clothes and furniture — all in one tool

Situation Analysis

Cat

Pet Hair on Furniture

A person warmly holding their pet cat — hair embeds deeply into sofa fabric, impossible to vacuum out completely.



Shirt

Hair on Clothes

Cat hair sticks on T-shirt while petting — clings to fabric fibres and cannot be brushed off easily by hand.



Basin

Fallen hairs collect on washbasins and floors after combing — builds up in corners and clogs drains.



Interview, Research Analysis & Insights

All 3 interviewees frequently notice hair stuck on clothes, carpets and sofas. Current methods: lint rollers, tape, broom, brush, vacuum cleaners.

Most find manual cleaning frustrating and unhygienic. All prefer a hands-free, reusable, compact, portable and pet-safe solution.

6 Thinking Hats

White Hat — Facts

Pet/human hair is one of the most common household cleaning complaints. Lint rollers are widely available. All interviewees find manual cleaning frustrating and unhygienic.



Yellow Hat — Optimism

Completely hands-free — never touch the hair. Musical feedback confirms the glove is working. Bristles + sticky palm collects hair in one motion.



Black Hat — Caution

Sticky pads generate recurring waste — sustainability concern. Electronics may be damaged by water exposure. Duck shape may feel uncomfortable for some hand sizes.



Green Hat — Creativity

Piano notes could play a full song when cleaning is complete. Lint pad could be made biodegradable. Kids version with animal sounds instead of piano.



Red Hat — Feelings

Feels like playing an instrument rather than a chore. Slight embarrassment of wearing it in public is very Chindogu and fun.



Blue Hat — Process

Prototype should separate mechanical and electronics testing. Test bristle firmness on multiple fabric types. Test on sofa, clothes and floor — at least 3 surfaces.



What It Is

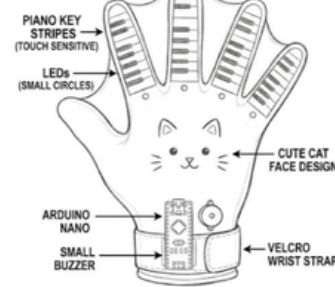
Right-handed smart glove combining rubber-bristled scrubbing with a sticky lint pad. Powered by an Arduino musical interface — turns cleaning into a rhythmic, playful experience.

The Chindogu Twist

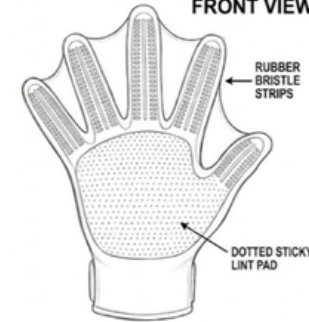
Every rub triggers an RGB LED piano key and plays a buzzer note — so cleaning your sofa accidentally turns you into a musician. Cat face on the back for maximum absurdity.

PRODUCT DESIGN: "DUCKY-CLEAN" ELECTRONIC GLOVE (RIGHT HAND ONLY)

BACK VIEW



FRONT VIEW



BACK VIEW



FRONT VIEW



STEP 8 — BUILD

Parts Required

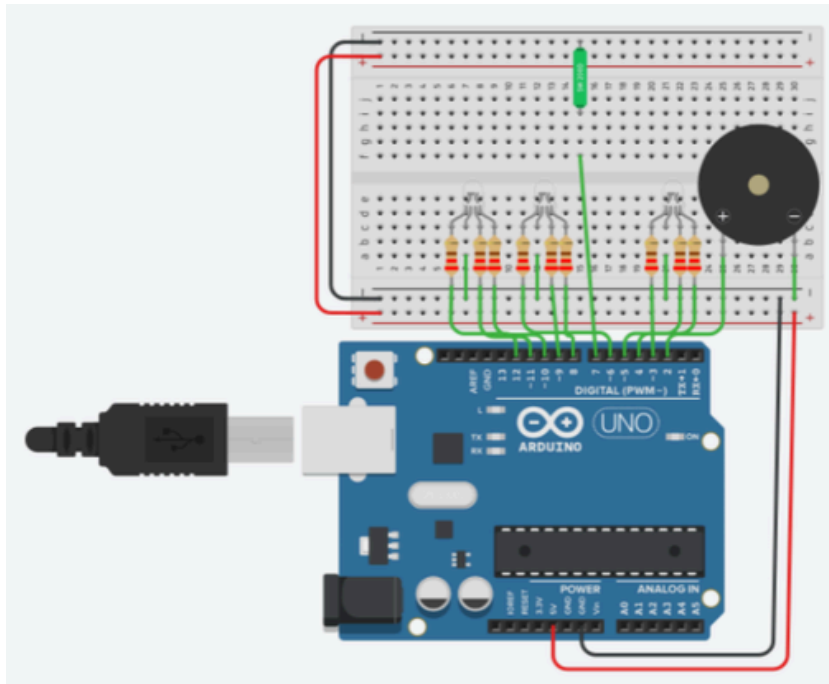
Mechanical Parts

- Rubber glove — base structure (right hand)
- Rubber bristles — attached along all finger surfaces
- Sticky lint roller pad — fixed to the palm area
- Velcro wrist strap — adjustable wrist closure

Electronics

- Arduino Nano — main microcontroller
- MPU-6050 — accelerometer/gyroscope (rub detection)
- RGB LEDs x3 — light up like piano keys on each rub
- Small Buzzer — plays musical notes per motion
- LiPo Battery — rechargeable wireless power supply

Tinkercad work



How it Works



Poster

QuackTrack

Clean the Mess. Play the Notes.
Look Ridiculous. Feel Great.



**Loosens
Stuck Hair**



**Collects
Without
Touching**



**Plays Music
While You
Clean**

Arduino Powered

estimated cost ~₹900 to build

QuackTrack — Mission Accomplished

Real Problem Solved

QuackTrack tackles a genuine, frustrating daily issue — pet and human hair stuck on sofas, clothes and floors that vacuums simply cannot reach.

Chindogu at its Finest

The musical glove with a cat face is delightfully absurd. It works — but wearing it in public would cause the exact mild social embarrassment Chindogu demands.

Full Design Process

From user research and interviews to 6 Thinking Hats ideation, prototyping, and Arduino simulation — every step of the design thinking process was applied.

User Research

Ideation

Prototyping

Arduino

Chindogu

Interaction Design